

Vasu Mekala

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EDUCATION

University of New Haven - Master of Science in Data Science

West Haven, CT

- **Relevant Coursework:** NLP, Deep Learning, Machine Learning, AI, Power BI

Hindustan Institute of Technology & Sciences - Bachelor of Technology in Information Technology

Chennai, India

- **Relevant Coursework:** Analytical Mathematics, Database Technologies, Decision Modeling, Predictive Analytics, Data Visualization, Machine Learning, AI, Business Intelligence

PROFESSIONAL CERTIFICATIONS

HackerRank - Intermediate SQL Certification

Mountain View, CA

IBM - SPSS Statistics Essential Training Certification

Armonk, NY

TECHNICAL SKILLS

Python, C, Java, SQL, Power BI, Azure ML Studio, PowerApps, MongoDB, MySQL, AWS (S3, EC2), GitHub, Excel, Steamlit, NLP, Deep Learning

WORK EXPERIENCE

HCLTech

Noida, Uttar Pradesh

Data Analyst

September 2022 - August 2023

- **Low-Code App Development:** Led a team of 3 in designing and deploying low-code business applications using Microsoft Power Apps, seamlessly integrating data from SharePoint, SQL Server, and Dataverse. This initiative improved data accessibility and boosted operational efficiency by 30%.
- **Workflow Automation:** Collaborated with cross-functional teams to identify repetitive tasks and implemented automated workflows in Power Automate, successfully reducing manual processing time by 40% and improving overall productivity.
- **Dashboard & Reporting:** Developed interactive, real-time dashboards using Power BI to visualize performance metrics and KPIs, empowering clients with actionable insights and reducing report generation time by 25%.

Wipro

Bengaluru, India

Project Engineer Trainee

February 2022 - August 2022

- **Real-Time Application Development:** Collaborated in a 6-member team to build a real-time COVID-19 case-tracking application using Python, SQL, and machine learning to support public health analytics.
- **API & Data Pipeline Engineering:** Spearheaded the development of API-driven data pipelines using Spring Boot, resulting in a 35% improvement in data processing speed.
- **Front-End Collaboration:** Worked closely with UI/UX designers to enhance the application's user interface and experience, ensuring intuitive navigation and increased user engagement.

Exotrix E-commerce

Chennai, India

Data Analytics Intern

July 2021 - December 2021

- **Customer Data Analysis:** Conducted Exploratory Data Analysis (EDA) on customer datasets, uncovering key insights that contributed to a 12% increase in customer retention.
- **Machine Learning Implementation:** Developed and deployed basic machine learning models to enhance product recommendations, boosting recommendation accuracy by 18%.
- **SQL Query Optimization:** Optimized complex SQL queries, leading to a 15% improvement in database query performance and faster data retrieval for business reporting.

PROJECT EXPERIENCE

Medical Term to Human Language Translate System

January 20205 – May 2025

- **Data Pipeline & OCR Optimization:** Built a reproducible Python pipeline to preprocess scanned PDF medical reports, using EDA to fine-tune image transformations and boost OCR accuracy to 93%.
- **Biomedical NLP & Anomaly Detection:** Applied SciSpacy, regex, and statistical analysis to perform biomedical NER and detect anomalies against JSON-based clinical reference ranges.
- **LLM Integration & Chatbot Development:** Developed NLP-driven explanation and translation workflows using Mistral LLM, Marian MT, and Jinja2; created a RAG-based chatbot with a Flask UI, reducing clinician follow-up workload by 30%.

Spotify Data Analysis

August 2024 – December 2024

- **Data Collection & Cleaning:** Collected, cleaned, and preprocessed Spotify's 2023 dataset of 20,000+ tracks using Python libraries Pandas and NumPy, ensuring high data quality and consistency for analysis.
- **Exploratory Data Analysis:** Performed exploratory data analysis to identify trends in song popularity, genre distribution, and artist metrics, leveraging Matplotlib and Seaborn to produce clear visualizations that revealed key correlations such as impact on popularity.
- **Natural Language Processing & Reporting:** Applied natural language processing (NLP) techniques on song titles and lyrics to extract dominant themes and sentiment patterns, synthesizing insights into a detailed report that informed stakeholders on evolving music trends and audience preferences.

- **Data Modeling Leadership:** Led a team of 4 to design and implement both transactional and star-schema data models, significantly enhancing data organization and improving accessibility across departments by 40%.
Power BI Dashboard Design: Independently developed and deployed a variety of dynamic Power BI dashboards and reports, tailored to end-user needs, which streamlined data visualization processes and reduced report generation time by 30%.
- **Machine Learning Collaboration:** Partnered closely with data engineering teams to develop and refine machine learning models using ML.NET, boosting prediction accuracy by 25%, which directly supported more informed decision-making and predictive analytics.

- **Model Development:** Independently designed and implemented a natural language processing (NLP) classification model using Python and scikit-learn to detect offensive language in tweets, achieving 83% accuracy on the test dataset.
- **Model Optimization:** Applied advanced feature engineering techniques such as TF-IDF vectorization and conducted hyperparameter tuning using GridSearchCV, leading to a 17% improvement in model accuracy and robustness.